

Latimer diagrams are concise summaries of the standard reduction potentials associated with the electrochemical conversions of an element that can exist in various oxidation states. Latimer diagrams consist of a series of chemical species that are progressively reduced and arranged in order of decreasing oxidation number with respect to the element of interest. The oxidation number for the element of interest is often listed above or below each species, but in some cases these values are omitted. The standard reduction potential E° for each reduction is written above the corresponding arrow joining the species. Latimer diagrams for chromium and ruthenium under acidic conditions are shown in Figures 1 and 2, respectively.

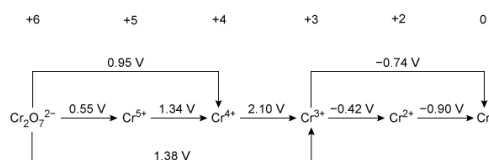


Figure 1 Latimer diagram for chromium in aqueous solution under acidic conditions (pH = 0)

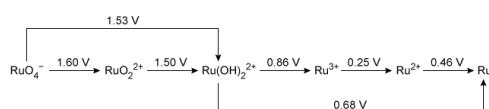


Figure 2 Latimer diagram for ruthenium in aqueous solution at pH = 0 (oxidation numbers not shown)

Latimer diagrams allow the redox properties of an element to be presented in a single line (rather than in several half-reactions) by omitting the aqueous acidic or basic species from the reactions and showing only the redox-active species. However, balanced half-reactions or net reactions can be easily written from a segment of a Latimer diagram by adding unstated electrons and either $\text{H}^+(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ (under acidic conditions) or $\text{OH}^-(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ (under basic conditions). Accordingly, Latimer diagrams can be used to evaluate hypothetical cells such as Figure 3.

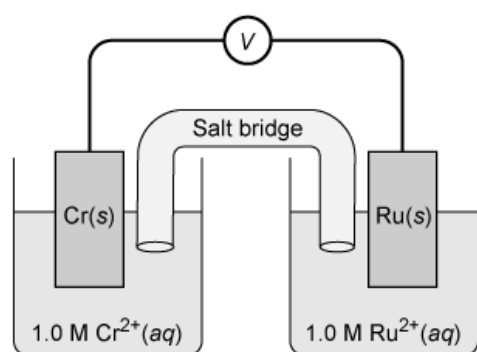


Figure 3 Hypothetical galvanic cell

For reactions involving more than one step on a Latimer diagram, the standard potentials between species are not additive. Accordingly, the overall potential $E^\circ_{\text{Steps } 1 \rightarrow n}$ required to convert one species to the other must be found by summing the standard change in Gibbs free energy ΔG° from each step and then dividing the sum by the product of Faraday's constant F and the overall change in the oxidation number ΔN between the species.

$$E^\circ_{\text{Steps } 1 \rightarrow n} = \frac{\Delta G^\circ_{\text{Step } 1} + \Delta G^\circ_{\text{Step } 2} + \dots + \Delta G^\circ_{\text{Step } n}}{\Delta NF}$$

Equation 1

In the $\text{Ru}(\text{OH})_2^{2+}$ cation, the Ru atom has an oxidation number of:

- ☐ A. 0 [6%]
☐ B. +2 [13%]
☒ C. +4 [75%]
☐ D. +6 [4%]

Correct

75% answered correctly

Explanation:

Sum of the oxidation numbers equals the net charge

Oxidation number (single atom):	+4	-2	+1
Cation formula:	$\text{Ru}(\text{OH})_2^{2+}$		
Totals (column):	+4	-4	+2
Total row must sum to the value of the net charge	$(+4) + (-4) + (+2) = +2$		

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Oxidation numbers account for the gain or loss of electrons by an atom relative to the number of electrons present in its elemental state. The oxidation number of an atom may be assessed from the compound formula using [oxidation state rules](#). By this method, atoms with consistent oxidation states (ie, those that rarely vary) are used to determine oxidation states for atoms that have variable states (eg, [transition metals](#)).

Oxidation numbers differ from the formal charge in that oxidation numbers account for electron changes due to [oxidation and reduction](#) whereas [formal charge](#) accounts for distribution of charge among atoms. However, in any molecule or ion the **sum of the oxidation states** of every atom in the structure must equal the **net charge** of the chemical formula.

Applying the **oxidation state rules** to the $\text{Ru}(\text{OH})_2^{2+}$ cation shows that the Ru atom has an oxidation state of +4. Although Ru is a transition metal that has a variable oxidation state (depending on the compound formed), the oxidation states of O and H are consistent (ie, -2 and +1, respectively) and can be used to evaluate the state of the Ru atom. Because the $\text{Ru}(\text{OH})_2^{2+}$ cation has a net charge of +2, the oxidation numbers of the Ru atom, the two H atoms (+1 each), and the two O atoms (-2 each) must sum to +2.

$$\text{Ru}(\text{OH})_2^{2+} = +2 \text{ (net charge)}$$

$$\text{Ru} + 2(-2) + 2(+1) = +2$$

Solving for Ru gives its oxidation number in the cation:

$$\text{Ru} - 4 + 2 = +2$$

$$\text{Ru} = (2 + 4 - 2) = +4$$

(Choice A) A value of 0 is the oxidation number of the Ru atom in its elemental state (ie, when the atom is not bonded to other elements and has not gained or lost electrons).

(Choice B) A value of +2 is the net charge of the cation.

(Choice D) A value of +6 is the oxidation number of the Ru atom in the RuO_2^{2+} cation, not the oxidation

number of the Ru atom in the $\text{Ru}(\text{OH})_2^{2+}$ cation.

Educational objective:

The oxidation number of each atom in a compound or ion formula may be determined using oxidation state rules and recognizing that the sum of the oxidation states of each atom must equal the net charge of the compound or ion.

Time spent:0 sec

Subject:
General Chemistry

QID:403851

System:
Solutions and Electrochemistry

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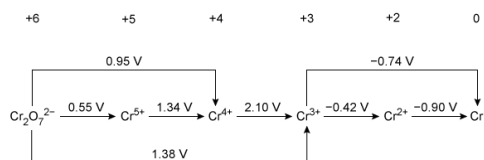


Figure 1 Latimer diagram for chromium in aqueous solution under acidic conditions (pH = 0)

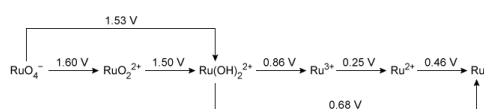


Figure 2 Latimer diagram for ruthenium in aqueous solution at pH = 0 (oxidation numbers not shown)

Latimer diagrams allow the redox properties of an element to be presented in a single line (rather than in several half-reactions) by omitting the aqueous acidic or basic species from the reactions and showing only the redox-active species. However, balanced half-reactions or net reactions can be easily written from a segment of a Latimer diagram by adding unstated electrons and either $\text{H}^+(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ (under acidic conditions) or $\text{OH}^-(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ (under basic conditions). Accordingly, Latimer diagrams can be used to evaluate hypothetical cells such as Figure 3.

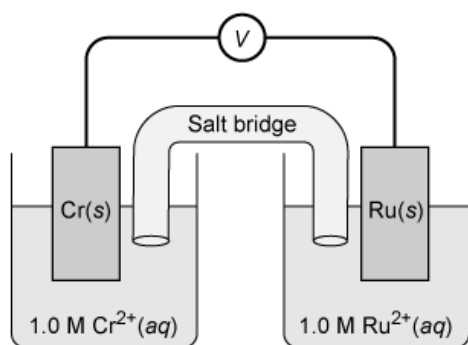


Figure 3 Hypothetical galvanic cell

For reactions involving more than one step on a Latimer diagram, the standard potentials between species are not additive. Accordingly, the overall potential $E^\circ_{\text{Steps } 1 \rightarrow n}$ required to convert one species to the other must be found by summing the standard change in Gibbs free energy ΔG° from each step and then dividing the sum by the product of Faraday's constant F and the overall change in the oxidation number ΔN between the species.

$$E^\circ_{\text{Steps } 1 \rightarrow n} = \frac{\Delta G^\circ_{\text{Step } 1} + \Delta G^\circ_{\text{Step } 2} + \dots + \Delta G^\circ_{\text{Step } n}}{\Delta N F}$$

Equation 1

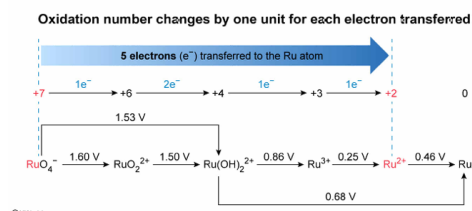
How many electrons are transferred to each ruthenium atom in a redox process that converts RuO_4^- into Ru^{2+} ?

- ☐ A. 2 electrons [8%]
☐ B. 3 electrons [15%]
☐ C. 4 electrons [17%]
✓ ☒ D. 5 electrons [58%]

Correct

59% answered correctly

Explanation:



In an electrochemical reaction, **oxidation and reduction** always occur simultaneously (ie, the electrons lost by the atom that is oxidized are gained by another atom that is reduced). To **track electron transfers** between atoms during redox reactions, an **oxidation number** (oxidation state) is assigned for each atom in a compound relative to the number of electrons in a neutral atom of that element.

Comparing the oxidation number of an element in the reactants with the oxidation number of that same element in the products indicates whether the element has gained or lost electrons. An increased oxidation number corresponds to a loss of electrons (oxidation) whereas a decreased oxidation number results from gaining electrons (reduction). A change of **one unit** occurs **for each electron** transferred. As such, the change in the oxidation number of an atom is equal to the number of electrons transferred to (or from) the atom during a reaction.

The conversion of RuO_4^- into Ru^{2+} involves the reduction of a Ru atom from an oxidation number of +7 in RuO_4^- to an oxidation number of +2 in Ru^{2+} (ie, a difference of 5 units). Therefore, five electrons are transferred to each Ru atom that is reduced.

(Choice A) Although the charge of the Ru^{2+} ion product is +2, this value does not indicate the total number of electrons transferred in the conversion.

(Choice B) Although the difference in the charge of the RuO_4^- and Ru^{2+} ions is 3 units (ie, $+2 - (-1) = +3$), the difference in ion charge does not always indicate the number of electrons transferred from a particular atom because the net charge of a polyatomic ion results from all contributing atoms.

(Choice C) Although the Latimer diagram includes four steps in the conversion of RuO_4^- into Ru^{2+} , not all steps involve one electron. The second step involves two electrons.

Educational objective:

Oxidation numbers track the transfer of electrons between atoms during a redox reaction. An increased oxidation number corresponds to a loss of electrons (oxidation) whereas a decreased oxidation number results from gaining electrons (reduction). A change of one unit occurs for each electron transferred.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403852

System:
Solutions and Electrochemistry

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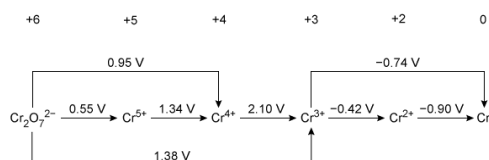


Figure 1 Latimer diagram for chromium in aqueous solution under acidic conditions (pH = 0)

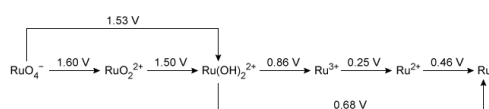


Figure 2 Latimer diagram for ruthenium in aqueous solution at pH = 0 (oxidation numbers not shown)

Latimer diagrams allow the redox properties of an element to be presented in a single line (rather than in several half-reactions) by omitting the aqueous acidic or basic species from the reactions and showing only the redox-active species. However, balanced half-reactions or net reactions can be easily written from a segment of a Latimer diagram by adding unstated electrons and either $\text{H}^+(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ (under acidic conditions) or $\text{OH}^-(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ (under basic conditions). Accordingly, Latimer diagrams can be used to evaluate hypothetical cells such as Figure 3.

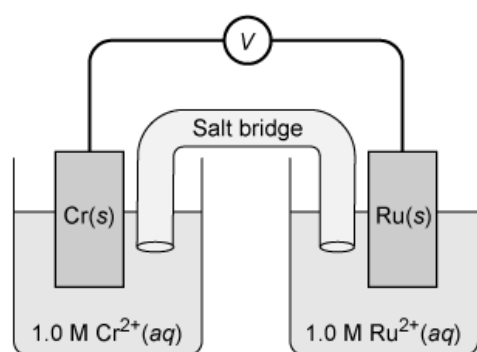


Figure 3 Hypothetical galvanic cell

For reactions involving more than one step on a Latimer diagram, the standard potentials between species are not additive. Accordingly, the overall potential $E^\circ_{\text{Steps } 1 \rightarrow n}$ required to convert one species to the other must be found by summing the standard change in Gibbs free energy ΔG° from each step and then dividing the sum by the product of Faraday's constant F and the overall change in the oxidation number ΔN between the species.

$$E^\circ_{\text{Steps } 1 \rightarrow n} = \frac{\Delta G^\circ_{\text{Step } 1} + \Delta G^\circ_{\text{Step } 2} + \dots + \Delta G^\circ_{\text{Step } n}}{\Delta NF}$$

Equation 1

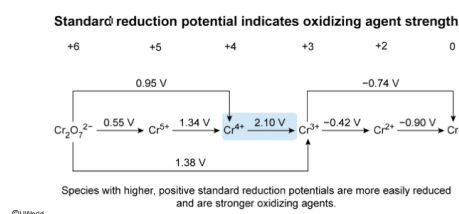
Based on the standard reduction potentials in Figure 1, which of the following chemical species is the strongest oxidizing agent?

- ☐ A. $\text{Cr}_2\text{O}_7^{2-}$ [23%]
☐ B. Cr^{5+} [5%]
☒ C. Cr^{4+} [47%]
☐ D. Cr^{3+} [23%]

Correct

46% answered correctly

Explanation:



In an **oxidation-reduction (redox) reaction**, **oxidation and reduction** always occur simultaneously. The electrons lost by the atom that is oxidized are gained by another atom that is reduced. As a result, the chemical species containing the atom being oxidized is the **reducing agent** that *causes reduction* in another atom by giving up the electrons involved in the reduction of the other atom. Conversely, the species with an atom that is reduced functions as the **oxidizing agent** that *causes oxidation* by taking electrons from (oxidizing) another atom.

A Latimer diagram gives a sequence of chemical reductions. The standard reduction potential (E°) above each arrow in the diagram indicates how readily the chemical species to the left of the arrow is reduced into the species to the right of the arrow. A **negative E°** indicates that the reduction is **unfavorable and nonspontaneous** whereas a **positive E°** indicates that the reduction is **favorable and spontaneous**. Consequently, species with **higher, positive E° values** are more easily reduced and are **stronger oxidizing agents** (ie, readily take electrons from other species).

Therefore, among the given choices, Cr^{4+} is the strongest oxidizing agent because it has the largest positive E° .

(Choices A and B) $\text{Cr}_2\text{O}_7^{2-}$ and Cr^{5+} are weaker oxidizing agents than Cr^{4+} because $\text{Cr}_2\text{O}_7^{2-}$ and Cr^{5+} have lower E° values than that of Cr^{4+} .

(Choice D) The negative E° value of Cr^{3+} indicates that its reduction is unfavorable. As such, the reverse reaction (ie, an oxidation) is more favorable, and Cr^{3+} is more likely to *lose electrons* and act as a *reducing agent*.

Educational objective:

A oxidizing agent is a chemical species that gets reduced and, in the process, causes oxidation in another species. Chemical species with higher, positive standard reduction potentials are more easily reduced and are stronger oxidizing agents than species with lower or negative standard reduction potentials.

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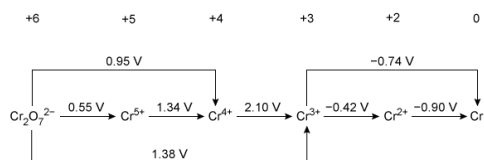


Figure 1 Latimer diagram for chromium in aqueous solution under acidic conditions (pH = 0)

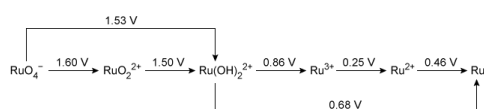


Figure 2 Latimer diagram for ruthenium in aqueous solution at pH = 0 (oxidation numbers not shown)

Latimer diagrams allow the redox properties of an element to be presented in a single line (rather than in several half-reactions) by omitting the aqueous acidic or basic species from the reactions and showing only the redox-active species. However, balanced half-reactions or net reactions can be easily written from a segment of a Latimer diagram by adding unstated electrons and either $H^+(aq)$ and $H_2O(l)$ (under acidic conditions) or $OH^-(aq)$ and $H_2O(l)$ (under basic conditions). Accordingly, Latimer diagrams can be used to evaluate hypothetical cells such as Figure 3.

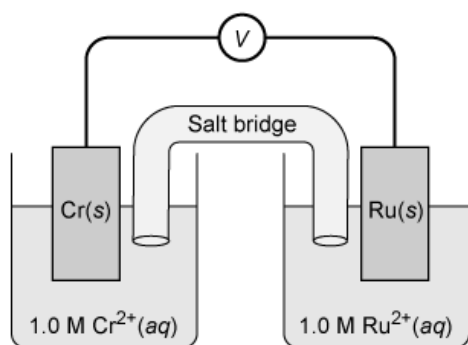


Figure 3 Hypothetical galvanic cell

For reactions involving more than one step on a Latimer diagram, the standard potentials between species are not additive. Accordingly, the overall potential $E^\circ_{\text{Steps } 1 \rightarrow n}$ required to convert one species to the other must be found by summing the standard change in Gibbs free energy ΔG° from each step and then dividing the sum by the product of Faraday's constant F and the overall change in the oxidation number ΔN between the species.

$$E^\circ_{\text{Steps } 1 \rightarrow n} = \frac{\Delta G^\circ_{\text{Step } 1} + \Delta G^\circ_{\text{Step } 2} + \dots + \Delta G^\circ_{\text{Step } n}}{\Delta NF}$$

Equation 1

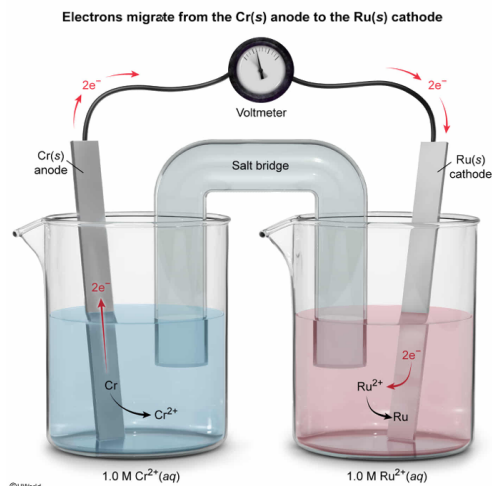
Which of the following correctly describes the migration of electrons during the operation of the cell in Figure 3? Electrons migrate from:

- ✓ ☒ A. Cr(s) to Ru(s) because Cr(s) is the anode. [62%]
☐ B. Cr(s) to Ru(s) because Cr(s) is the cathode. [12%]
☐ C. Ru(s) to Cr(s) because Ru(s) is the anode. [18%]
☐ D. Ru(s) to Cr(s) because Ru(s) is the cathode. [6%]

Correct

63% answered correctly

Explanation:



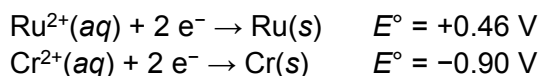
In any electrochemical cell, electrons are liberated at the **anode** (ie, the electrode where **oxidation** takes place) and transferred to the **cathode** (ie, the electrode where **reduction** occurs). The net redox reaction resulting from the electrode activity yields a standard cell potential E°_{cell} , which is calculated by:

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

where E°_{cathode} and E°_{anode} are the standard reduction potentials of the half-reactions occurring at the cathode and the anode, respectively.

A **positive** E°_{cell} indicates a thermodynamically **favorable reaction** suitable for a **galvanic cell** whereas a **negative** E°_{cell} indicates an **unfavorable reaction** that can only be achieved by an **electrolytic cell** in which external energy is supplied to the system. Consequently, the cell in Figure 3 can only function as a galvanic cell when its E°_{cell} is positive.

Taking the relevant half-reactions from the Latimer diagrams in Figures 1 and 2 for the cell in Figure 3 gives:



Comparing the E° values for these half-reactions shows that E°_{cell} is positive only if Cr(s) is the anode.

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$E^\circ_{\text{cell}} = +0.46 \text{ V} - (-0.90 \text{ V}) = +1.36 \text{ V}$$

Therefore, when the cell in Figure 3 operates as a galvanic cell, electrons are transferred from Cr(s) to Ru(s) because Cr(s) is the anode.

(Choices B and D) In any electrochemical cell, electrons flow from the *anode to the cathode*. If the Cr(s) or Ru(s) electrode functions as the cathode, the electrons would migrate toward (not from) it.

(Choice C) Although electrons could migrate from Ru(s) to Cr(s) if Ru(s) functioned as the anode, this cannot happen in a galvanic cell because the reaction is unfavorable (ie, E°_{cell} is negative).

$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$E^\circ_{\text{cell}} = -0.90 \text{ V} - (+0.46 \text{ V}) = -1.36 \text{ V}$$

Educational objective:

In an electrochemical cell, electrons are always transferred from the anode (where oxidation occurs) to the cathode (where reduction occurs). In a galvanic cell, the overall cell potential must be positive, and the electrode with the greater standard reduction potential has a greater tendency to be reduced and function as the cathode.

Time spent:0 sec

Subject:
General Chemistry

QID:403854

System:
Solutions and Electrochemistry

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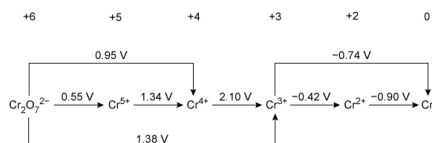


Figure 1 Latimer diagram for chromium in aqueous solution under acidic conditions (pH = 0)

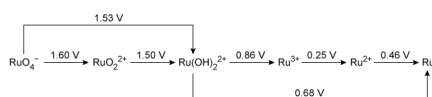


Figure 2 Latimer diagram for ruthenium in aqueous solution at pH = 0 (oxidation numbers not shown)

Latimer diagrams allow the redox properties of an element to be presented in a single line (rather than in several half-reactions) by omitting the aqueous acidic or basic species from the reactions and showing only the redox-active species. However, balanced half-reactions or net reactions can be easily written from a segment of a Latimer diagram by adding unstated electrons and either $\text{H}^+(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ (under acidic conditions) or $\text{OH}^-(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ (under basic conditions). Accordingly, Latimer diagrams can be used to evaluate hypothetical cells such as Figure 3.

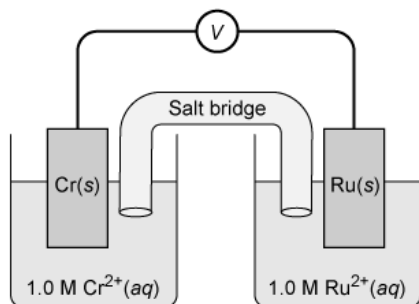


Figure 3 Hypothetical galvanic cell

For reactions involving more than one step on a Latimer diagram, the standard potentials between species are not additive. Accordingly, the overall potential $E^\circ_{\text{Steps } 1 \rightarrow n}$ required to convert one species to the other must be found by summing the standard change in Gibbs free energy ΔG° from each step and then dividing the sum by the product of Faraday's constant F and the overall change in the oxidation number ΔN between the species.

$$E^\circ_{\text{Steps } 1 \rightarrow n} = \frac{\Delta G^\circ_{\text{Step 1}} + \Delta G^\circ_{\text{Step 2}} + \dots + \Delta G^\circ_{\text{Step } n}}{\Delta N F}$$

Equation 1

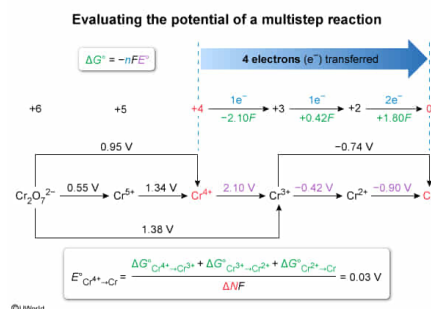
Which expression correctly evaluates the overall potential E° required for the multi-step conversion of Cr^{4+} to Cr ?

- ✓ ☒ A. $E^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}} = \frac{-2.10F + 0.42F + 1.80F}{(+4 - 0)F} = 0.03 \text{ V}$ [56%]
- ☐ B. $E^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}} = \frac{-2.10F + 0.42F + 0.90F}{(+3 - 0)F} = 0.26 \text{ V}$ [17%]
- ☐ C. $E^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}} = 2.10 \text{ V} - 0.42 \text{ V} - 0.90 \text{ V} = 0.78 \text{ V}$ [12%]
- ☐ D. $E^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}} = 2.10 \text{ V} - 0.74 \text{ V} = 1.36 \text{ V}$ [13%]

Correct

57% answered correctly

Explanation:



The change in the standard Gibbs free energy ΔG° associated with an oxidation-reduction (redox) reaction is proportional to the standard potential E° of the reaction according to the equation

$$\Delta G^\circ = -nFE^\circ$$

where n is the number of electrons transferred per reaction and F is Faraday's constant. As such, a positive E° indicates a thermodynamically favorable reaction whereas a negative E° indicates an unfavorable reaction in which external energy must be supplied to the system.

The Latimer diagram in Figure 1 shows that the conversion of Cr^{4+} to Cr involves three steps. Accordingly, the ΔG° for each step is:

$$\Delta G^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}^{3+}} = -(1 e^-) \left(F \frac{\text{J}}{\text{V} \cdot \text{mol } e^-} \right) (+2.10 \text{ V}) = -2.10F \frac{\text{J}}{\text{mol}}$$

$$\Delta G^\circ_{\text{Cr}^{3+} \rightarrow \text{Cr}^{2+}} = -(1 e^-) \left(F \frac{\text{J}}{\text{V} \cdot \text{mol } e^-} \right) (-0.42 \text{ V}) = +0.42F \frac{\text{J}}{\text{mol}}$$

$$\Delta G^\circ_{\text{Cr}^{2+} \rightarrow \text{Cr}} = -(2 e^-) \left(F \frac{\text{J}}{\text{V} \cdot \text{mol } e^-} \right) (-0.90 \text{ V}) = +1.80F \frac{\text{J}}{\text{mol}}$$

In each step, n is equal to the change in the oxidation number N shown above the species on the Latimer diagram. According to Equation 1, the E° required for the conversion of Cr^{4+} to Cr is calculated as:

$$E^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}} = \frac{\Delta G^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}^{3+}} + \Delta G^\circ_{\text{Cr}^{3+} \rightarrow \text{Cr}^{2+}} + \Delta G^\circ_{\text{Cr}^{2+} \rightarrow \text{Cr}}}{\Delta n F}$$

$$E^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}} = \frac{-2.10F + 0.42F + 1.80F}{(+4 - 0)F} = \frac{0.12F}{4F} = 0.03 \text{ V}$$

Alternatively, because the Latimer diagram also gives the overall E° value for the last two steps, the answer can be found more concisely by:

$$\Delta G^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}^{3+}} = -(1 e^-) \left(F \frac{\text{J}}{\text{V} \cdot \text{mol } e^-} \right) (+2.10 \text{ V}) = -2.10F \frac{\text{J}}{\text{mol}}$$

$$\Delta G^\circ_{\text{Cr}^{3+} \rightarrow \text{Cr}} = -(3 e^-) \left(F \frac{\text{J}}{\text{V} \cdot \text{mol } e^-} \right) (-0.74 \text{ V}) = +2.22F \frac{\text{J}}{\text{mol}}$$

which yields the same E° required for the conversion of Cr^{4+} to Cr :

$$E^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}} = \frac{\Delta G^\circ_{\text{Cr}^{4+} \rightarrow \text{Cr}^{3+}} + \Delta G^\circ_{\text{Cr}^{3+} \rightarrow \text{Cr}}}{\Delta n F} = \frac{-2.10F + 2.22F}{(+4 - 0)F} = \frac{0.12F}{4F} = 0.03 \text{ V}$$

(Choice B) A value of 0.26 V is obtained by mistakenly assessing a transfer of one electron (ie, $n = 1$) for the $\text{Cr}^{2+} \rightarrow \text{Cr}$ conversion step and subsequently applying an incorrect overall transfer of $\Delta n = 3$.

(Choices C and D) Values of 0.78 V and 1.36 V are found by adding the potentials listed between steps along two different paths between the Cr^{4+} and Cr species on the Latimer diagram.

$$\text{Path 1: } +2.10 \text{ V} - 0.42 \text{ V} - 0.90 \text{ V} = 0.78 \text{ V}$$

$$\text{Path 2: } +2.10 \text{ V} - 0.74 \text{ V} = 1.36 \text{ V}$$

However, the passage states that for reactions involving more than one step on a Latimer diagram, the standard potentials between the species are not additive, and the overall reduction potential must be derived from the changes in Gibbs free energy associated with each step according to Equation 1.

Educational objective:

The change in the standard Gibbs free energy ΔG° associated with an oxidation-reduction reaction is proportional to the standard potential E° of the reaction according to the equation $\Delta G^\circ = -nFE^\circ$, where n is the number of electrons transferred in the reaction, E° is the standard potential, and F is Faraday's constant.

Time spent:0 sec

Subject:
General Chemistry

QID:403855

System:
Solutions and Electrochemistry

Latimer diagrams are concise summaries of the standard reduction potentials associated with the electrochemical conversions of an element that can exist in various oxidation states. Latimer diagrams consist of a series of chemical species that are progressively reduced and arranged in order of decreasing oxidation number with respect to the element of interest. The oxidation number for the element of interest is often listed above or below each species, but in some cases these values are omitted. The standard reduction potential E° for each reduction is written above the corresponding arrow joining the species. Latimer diagrams for chromium and ruthenium under acidic conditions are shown in Figures 1 and 2, respectively.

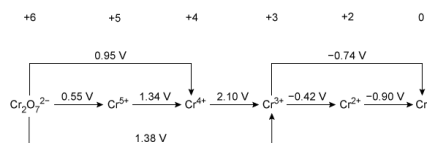


Figure 1 Latimer diagram for chromium in aqueous solution under acidic conditions (pH = 0)

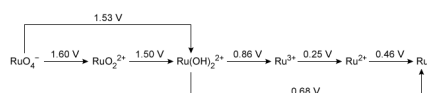


Figure 2 Latimer diagram for ruthenium in aqueous solution at pH = 0 (oxidation numbers not shown)

Latimer diagrams allow the redox properties of an element to be presented in a single line (rather than in several half-reactions) by omitting the aqueous acidic or basic species from the reactions and showing only the redox-active species. However, balanced half-reactions or net reactions can be easily written from a segment of a Latimer diagram by adding unstated electrons and either $\text{H}^+(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ (under acidic conditions) or $\text{OH}^-(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ (under basic conditions). Accordingly, Latimer diagrams can be used to evaluate hypothetical cells such as Figure 3.

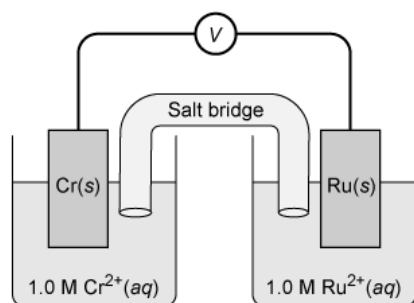


Figure 3 Hypothetical galvanic cell

For reactions involving more than one step on a Latimer diagram, the standard potentials between species are not additive. Accordingly, the overall potential $E^\circ_{\text{Steps } 1 \rightarrow n}$ required to convert one species to the other must be found by summing the standard change in Gibbs free energy ΔG° from each step and then dividing the sum by the product of Faraday's constant F and the overall change in the oxidation number ΔN between the species.

$$E^\circ_{\text{Steps } 1 \rightarrow n} = \frac{\Delta G^\circ_{\text{Step 1}} + \Delta G^\circ_{\text{Step 2}} + \dots + \Delta G^\circ_{\text{Step } n}}{\Delta N F}$$

Equation 1

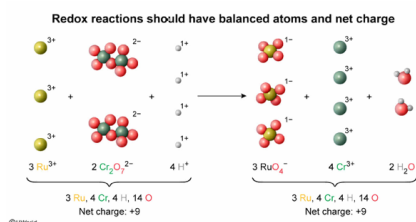
Suppose that $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$ undergoes an oxidation-reduction reaction with $\text{Ru}^{3+}(\text{aq})$ under acidic conditions and forms a species in which the ruthenium atom has an oxidation number of +7. Which of the following gives the balanced chemical equation for the reaction?

- ☐ A. $6 \text{Ru}^{3+}(\text{aq}) + \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 7 \text{H}_2\text{O}(\text{l}) \rightarrow 6 \text{Ru}(\text{OH})_2^{2+}(\text{aq}) + 2 \text{Cr}^{3+}(\text{aq}) + 2 \text{OH}^-(\text{aq})$ [7%]
- ☐ B. $\text{Ru}^{3+}(\text{aq}) + 2 \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 28 \text{H}^+(\text{aq}) \rightarrow \text{Ru}^{2+}(\text{aq}) + 4 \text{Cr}^{3+}(\text{aq}) + 14 \text{H}_2\text{O}(\text{l})$ [5%]
- ☐ C. $2 \text{Ru}^{3+}(\text{aq}) + 2 \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14 \text{H}^+(\text{aq}) \rightarrow 2 \text{RuO}_2^{2+}(\text{aq}) + 4 \text{Cr}^{3+}(\text{aq}) + 7 \text{H}_2\text{O}(\text{l})$ [13%]
- ☒ D. $3 \text{Ru}^{3+}(\text{aq}) + 2 \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 4 \text{H}^+(\text{aq}) \rightarrow 3 \text{RuO}_4^-(\text{aq}) + 4 \text{Cr}^{3+}(\text{aq}) + 2 \text{H}_2\text{O}(\text{l})$ [73%]

Correct

73% answered correctly

Explanation:



Redox reactions are fully balanced when the same number of each type of atom and the same net charge are present on both sides of the reaction equation. As such, the correct answer to the question can be found most simply by identifying the unbalanced reaction choices or those that do not match the reaction description. However, the correct (balanced) equation can be determined and verified more rigorously using half-reactions.

Oxidation-reduction (redox) reactions consist of two simultaneous **half-reactions** (one for the **oxidation** and one for the **reduction**) that add together to give the overall reaction. In a balanced redox reaction, the total number of electrons released from the oxidation half-reaction must equal the total number of electrons used by the reduction half-reaction. Consequently, the first step in balancing a redox reaction is to balance the number of electrons exchanged.

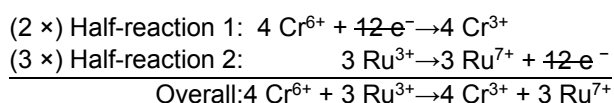
In the given reaction involving $\text{Cr}_2\text{O}_7^{2-}$, both **Cr atoms are reduced** from an **oxidation state** of +6 to a state of +3 as Cr^{3+} ions (a gain of three electrons per Cr atom and six electrons total).

Half-reaction 1 (reduction): $2 \text{ Cr}^{6+} + 6 \text{ e}^- \rightarrow 2 \text{ Cr}^{3+}$

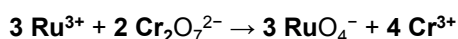
Simultaneously, **Ru atoms are oxidized** from an **oxidation state** of +3 in the Ru^{3+} ions to a state of +7 in the RuO_4^- ions (ie, a loss of four electrons per Ru atom).

Half-reaction 2 (oxidation): $\text{Ru}^{3+} \rightarrow \text{Ru}^{7+} + 4 \text{ e}^-$

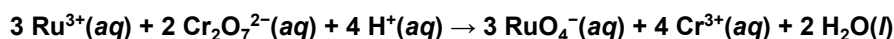
To balance and **cancel the electron terms** in the **overall reaction**, **multiples** of both half-reactions can be added together.



Based on the electron balance above, the coefficients of the Cr and Ru ions in the balanced reaction must be:



The next step is to **balance the atoms and the net charge**. Because the reaction occurs in an **aqueous solution** under **acidic conditions**, H_2O and H^+ can be added as reactants or products to complete the balancing of the equation. Therefore, the fully balanced redox reaction is:



(Choice A) This equation describes basic conditions (ie, OH^- is formed as a product). The question states that the reaction occurs under *acidic conditions*.

(Choice B) The net charge is not balanced in this equation (ie, +27 for the reactants, +14 for the products).

(Choice C) The atoms are not balanced in this equation (ie, 14 O atoms in the reactants, 11 O atoms in the products).

Educational objective:

Redox reactions are fully balanced when the same number of each type of atom and the same net charge are present on both sides of the reaction equation. For redox processes expressed as two half-reactions, the net reaction is found by adding the two half-reactions (or multiples of them) together so that the electron terms cancel.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403856

System:
Solutions and Electrochemistry

Researchers investigated the role of an electrochemically enhanced sulfate radical-based system in the inactivation of antibiotic-resistant *Escherichia coli* bacteria. An electrochemical cell (Figure 1) was constructed using an iron electrode and a graphite electrode connected by a copper wire joined to a voltmeter. Both electrodes were placed in a beaker containing a 3 mM solution of $\text{Na}_2\text{SO}_4(\text{aq})$ and a 50 mM solution of a peroxydisulfate salt, $\text{Na}_2\text{S}_2\text{O}_8(\text{aq})$. During the operation of the cell, $\text{Fe}^{2+}(\text{aq})$ was formed at the iron electrode and subsequently oxidized to $\text{Fe}^{3+}(\text{aq})$ by a reaction with $\text{S}_2\text{O}_8^{2-}(\text{aq})$. At the graphite electrode, $\text{Fe}^{2+}(\text{aq})$ was regenerated.

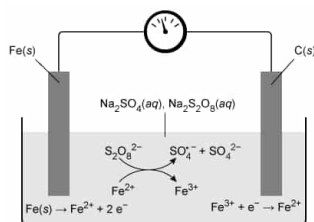
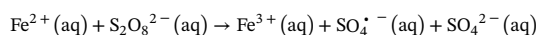


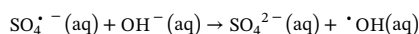
Figure 1 Electrochemical cell utilized in the inactivation of antibiotic-resistant *E. coli*

The reactions in the cell also produced both sulfate and hydroxyl radicals (ie, species with an unpaired electron). Sulfate radicals ($\text{SO}_4^{\bullet -}$) inactivate *E. coli* through membrane destruction and were produced in the electrochemical cell by Reaction 1.



Reaction 1

Operation of the cell generated enough energy to cause the electrolysis of water in which water was oxidized at one electrode to form $\text{H}^+(\text{aq})$ and reduced at the other electrode to form $\text{OH}^-(\text{aq})$. Consequently, some of the available $\text{OH}^-(\text{aq})$ reacted with $\text{SO}_4^{\bullet -}(\text{aq})$ to form hydroxyl radicals ($\text{OH}^{\bullet}(\text{aq})$), which also aided in the inactivation of *E. coli*.



Reaction 2

To determine which radical species predominately inactivated *E. coli* in the cell, two radical scavengers, methanol and *tert*-butanol, were used to evaluate the contributions of $\text{OH}^{\bullet}(\text{aq})$ and $\text{SO}_4^{\bullet -}(\text{aq})$. Methanol consumes both $\text{OH}^{\bullet}(\text{aq})$ and $\text{SO}_4^{\bullet -}(\text{aq})$ whereas *tert*-butanol primarily consumes $\text{OH}^{\bullet}(\text{aq})$. The researchers performed a series of experiments in which the concentration of *E. coli* was monitored in the active cell with and without the radical scavengers as a function of time. The results are shown in Figure 2.

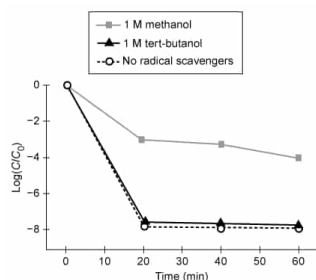


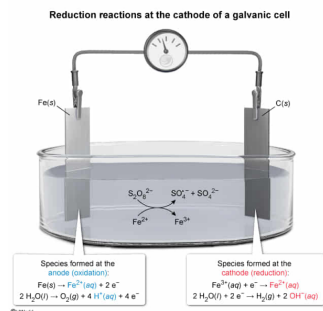
Figure 2 The effect of radical scavengers on the inactivation of antibiotic-resistant *E. coli* (Note: C_0 is the initial concentration of *E. coli*, and C is the concentration of *E. coli* at a specific time during the experiment.)

Which of the following species are formed at the cathode?

- I. $\text{Fe}^{3+}(\text{aq})$
 - II. $\text{OH}^-(\text{aq})$
 - III. $\text{Fe}^{2+}(\text{aq})$
 - IV. $\text{H}^+(\text{aq})$
- ☐ A. I and II only [7%]
☒ B. II and III only [79%]
☐ C. I, II, and IV only [3%]
☐ D. II, III, and IV only [9%]

Correct
78% answered correctly

Explanation:



In a **galvanic cell**, an **oxidation-reduction** (redox) reaction occurs **spontaneously** (ie, an external power source is not needed to drive the reaction), producing a flow of electrons between two electrodes connected by a wire. Electrons are generated at the **anode** (ie, the electrode where **oxidation** occurs) and flow through the wire toward the **cathode** (ie, the electrode where **reduction** occurs).

The galvanic cell described in the passage is composed of Fe(s) and C(s) electrodes. According to Figure 1, Fe(s) atoms are oxidized to Fe²⁺(aq) at the Fe(s) electrode, making the Fe(s) electrode the anode. During this process, water is simultaneously oxidized to H⁺(aq) (and O₂ gas) at the Fe(s) surface.

The electrons liberated at the anode flow through the wire to the inert C(s) electrode (cathode), where Fe³⁺ (formed via Reaction 1) is reduced back to Fe²⁺. As this occurs, water is simultaneously reduced to OH⁻(aq) (and H₂ gas) at the C(s) surface. Therefore, both Fe²⁺(aq) and OH⁻(aq) are formed at the C(s) cathode **(Numbers II and III)**.

(Numbers I and IV) Both Fe³⁺(aq) and H⁺(aq) are formed at the Fe(s) *anode*, where oxidation occurs.

Educational objective:

In an electrochemical cell, chemical species are oxidized at the anode and reduced at the cathode.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403857

System:
Solutions and Electrochemistry

Researchers investigated the role of an electrochemically enhanced sulfate radical-based system in the inactivation of antibiotic-resistant *Escherichia coli* bacteria. An electrochemical cell (Figure 1) was constructed using an iron electrode and a graphite electrode connected by a copper wire joined to a voltmeter. Both electrodes were placed in a beaker containing a 3 mM solution of $\text{Na}_2\text{SO}_4(\text{aq})$ and a 50 mM solution of a peroxydisulfate salt, $\text{Na}_2\text{S}_2\text{O}_8(\text{aq})$. During the operation of the cell, $\text{Fe}^{2+}(\text{aq})$ was formed at the iron electrode and subsequently oxidized to $\text{Fe}^{3+}(\text{aq})$ by a reaction with $\text{S}_2\text{O}_8^{2-}(\text{aq})$. At the graphite electrode, $\text{Fe}^{2+}(\text{aq})$ was regenerated.

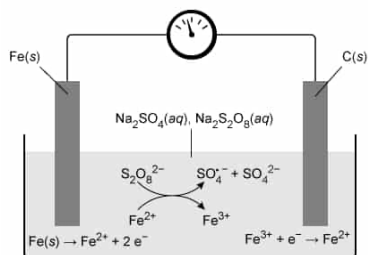
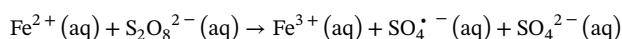


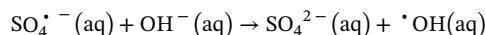
Figure 1 Electrochemical cell utilized in the inactivation of antibiotic-resistant *E. coli*

The reactions in the cell also produced both sulfate and hydroxyl radicals (ie, species with an unpaired electron). Sulfate radicals ($\text{SO}_4^{\bullet -}$) inactivate *E. coli* through membrane destruction and were produced in the electrochemical cell by Reaction 1.



Reaction 1

Operation of the cell generated enough energy to cause the electrolysis of water in which water was oxidized at one electrode to form $\text{H}^+(\text{aq})$ and reduced at the other electrode to form $\text{OH}^-(\text{aq})$. Consequently, some of the available $\text{OH}^-(\text{aq})$ reacted with $\text{SO}_4^{\bullet -}(\text{aq})$ to form hydroxyl radicals ($\text{OH}^{\bullet}$), which also aided in the inactivation of *E. coli*.



Reaction 2

To determine which radical species predominately inactivated *E. coli* in the cell, two radical scavengers, methanol and *tert*-butanol, were used to evaluate the contributions of $\text{OH}^{\bullet}(\text{aq})$ and $\text{SO}_4^{\bullet -}(\text{aq})$. Methanol consumes both $\text{OH}^{\bullet}(\text{aq})$ and $\text{SO}_4^{\bullet -}(\text{aq})$ whereas *tert*-butanol primarily consumes $\text{OH}^{\bullet}(\text{aq})$. The researchers performed a series of experiments in which the concentration of *E. coli* was monitored in the active cell with and without the radical scavengers as a function of time. The results are shown in Figure 2.

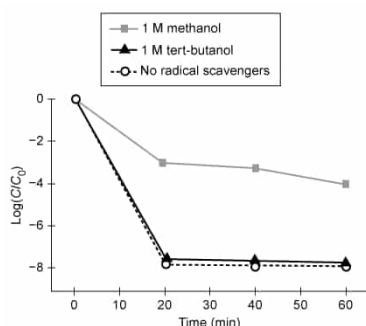


Figure 2 The effect of radical scavengers on the inactivation of antibiotic-resistant *E. coli* (Note: C_0 is the initial concentration of *E. coli*, and C is the concentration of *E. coli* at a specific time during the experiment.)

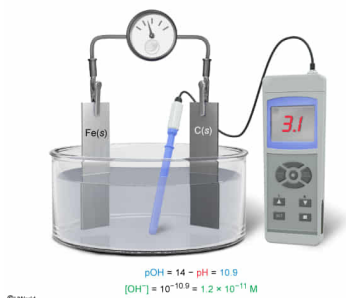
After the electrochemical cell in Figure 1 had operated for 60 minutes, the pH of the resulting solution was measured to be 3.1. What was the concentration of $\text{OH}^-(\text{aq})$ in the solution at that time?

- ☐ A. $7.9 \times 10^{-18} \text{ M}$ [3%]
- ☐ B. $1.0 \times 10^{-14} \text{ M}$ [5%]
- ☒ C. $1.2 \times 10^{-11} \text{ M}$ [83%]
- ☐ D. $7.9 \times 10^{-4} \text{ M}$ [7%]

Correct
84% answered correctly

Explanation:

Calculating the concentration of OH^- given the pH of a solution



The **pH scale** indicates the acidity of a solution, and a **pH value** is directly related to the **molar concentration** of $\text{H}^+(\text{aq})$ in the solution:

$$\text{pH} = -\log[\text{H}^+]$$

In contrast, the **pOH scale** indicates the basicity of a solution, and the **pOH value** is directly related to the molar concentration of $\text{OH}^-(\text{aq})$ in the solution:

$$\text{pOH} = -\log[\text{OH}^-]$$

The pH and pOH of a solution are interrelated (as derived from the self-ionization constant K_w of water) according to the equation:

$$\text{pH} + \text{pOH} = 14$$

The question states that after the electrochemical cell has operated for 60 minutes, the pH of the solution is measured to be 3.1. Subtracting the pH value from 14 yields the pOH of the solution:

$$\text{pOH} = 14 - 3.1 = 10.9$$

Therefore, taking the inverse logarithm (base 10) of both sides of the pOH equation and solving for $[\text{OH}^-]$ yields the molar concentration of $\text{OH}^-(\text{aq})$ present in the solution:

$$\text{pOH} = -\log[\text{OH}^-]$$

$$10^{-\text{pOH}} = 10^{\log[\text{OH}^-]}$$

$$10^{-\text{pOH}} = [\text{OH}^-]$$

$$[\text{OH}^-] = 10^{-10.9} = 1.2 \times 10^{-11} \text{ M}$$

Without a calculator, the correct answer can be identified from among the given choices by recognizing that the exact value must be close to $1 \times 10^{-11} \text{ M}$.

(Choice A) If the pH is *added* to 14 rather than *subtracted* from 14, a value of $7.9 \times 10^{-18} \text{ M}$ is obtained.

(Choice B) A value of $1.0 \times 10^{-14} \text{ M}$ is obtained if $\text{pOH} = 14$ is used in the calculation. A pOH of 14 would require that $\text{pH} = 0$.

(Choice D) A value of $7.9 \times 10^{-4} \text{ M}$ corresponds to the concentration of $\text{H}^+(\text{aq})$ (ie, $[\text{H}^+] = 10^{-\text{pH}}$), not the concentration of $\text{OH}^-(\text{aq})$.

Educational objective:

The pH of a solution is related to the OH^- concentration by the relationship $\text{pH} + \text{pOH} = 14$, where $\text{pOH} = -\log[\text{OH}^-]$.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403858

System:
Solutions and Electrochemistry

Researchers investigated the role of an electrochemically enhanced sulfate radical-based system in the inactivation of antibiotic-resistant *Escherichia coli* bacteria. An electrochemical cell (Figure 1) was constructed using an iron electrode and a graphite electrode connected by a copper wire joined to a voltmeter. Both electrodes were placed in a beaker containing a 3 mM solution of $\text{Na}_2\text{SO}_4(\text{aq})$ and a 50 mM solution of a peroxydisulfate salt, $\text{Na}_2\text{S}_2\text{O}_8(\text{aq})$. During the operation of the cell, $\text{Fe}^{2+}(\text{aq})$ was formed at the iron electrode and subsequently oxidized to $\text{Fe}^{3+}(\text{aq})$ by a reaction with $\text{S}_2\text{O}_8^{2-}(\text{aq})$. At the graphite electrode, $\text{Fe}^{2+}(\text{aq})$ was regenerated.

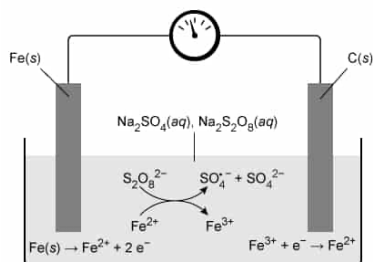
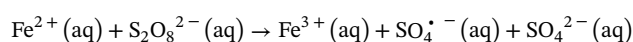


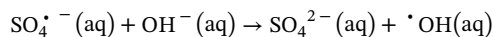
Figure 1 Electrochemical cell utilized in the inactivation of antibiotic-resistant *E. coli*

The reactions in the cell also produced both sulfate and hydroxyl radicals (ie, species with an unpaired electron). Sulfate radicals ($\text{SO}_4^{\bullet -}$) inactivate *E. coli* through membrane destruction and were produced in the electrochemical cell by Reaction 1.



Reaction 1

Operation of the cell generated enough energy to cause the electrolysis of water in which water was oxidized at one electrode to form $\text{H}^+(\text{aq})$ and reduced at the other electrode to form $\text{OH}^-(\text{aq})$. Consequently, some of the available $\text{OH}^-(\text{aq})$ reacted with $\text{SO}_4^{\bullet -}(\text{aq})$ to form hydroxyl radicals ($\text{OH}^{\bullet}$), which also aided in the inactivation of *E. coli*.



Reaction 2

To determine which radical species predominately inactivated *E. coli* in the cell, two radical scavengers, methanol and *tert*-butanol, were used to evaluate the contributions of $\text{OH}^{\bullet}(\text{aq})$ and $\text{SO}_4^{\bullet -}(\text{aq})$. Methanol consumes both $\text{OH}^{\bullet}(\text{aq})$ and $\text{SO}_4^{\bullet -}(\text{aq})$ whereas *tert*-butanol primarily consumes $\text{OH}^{\bullet}(\text{aq})$. The researchers performed a series of experiments in which the concentration of *E. coli* was monitored in the active cell with and without the radical scavengers as a function of time. The results are shown in Figure 2.

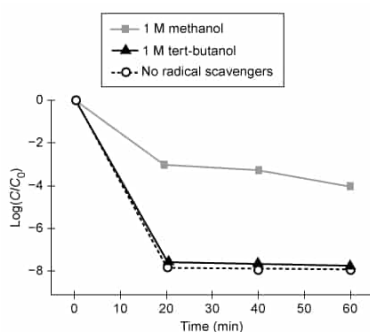


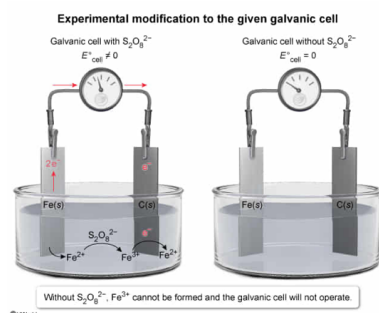
Figure 2 The effect of radical scavengers on the inactivation of antibiotic-resistant *E. coli* (Note: C_0 is the initial concentration of *E. coli*, and C is the concentration of *E. coli* at a specific time during the experiment.)

Which experimental modification to the electrochemical cell would result in a voltmeter reading of 0 V?

- ☒ A. Not adding the 50 mM $\text{Na}_2\text{S}_2\text{O}_8(\text{aq})$ [59%]
- ☐ B. Replacing the 50 mM $\text{Na}_2\text{S}_2\text{O}_8(\text{aq})$ with 50 mM $\text{FeCl}_3(\text{aq})$ [24%]
- ☐ C. Connecting the electrodes with a platinum wire [11%]
- ☐ D. Replacing the 3 mM $\text{Na}_2\text{SO}_4(\text{aq})$ with 3 mM $\text{MgSO}_4(\text{aq})$ [4%]

Correct
60% answered correctly

Explanation:



A typical **galvanic cell** consists of two metal electrodes (ie, **an anode and a cathode**), each immersed in separate solutions containing corresponding cations and connected by a metallic conducting wire. At the anode, one reactant loses electrons (**oxidation**), which then flow through the connecting wire to the cathode, where another reactant gains the lost electrons (reduction). This produces an electric current that can be used to do work, which is inferred from the **standard cell potential** E°_{cell} (ie, the voltage difference between the anode and cathode as measured by a voltmeter). If no current is flowing through the cell, $E^\circ_{\text{cell}} = 0 \text{ V}$.

In contrast, the galvanic cell described in the passage consists of a metal electrode (iron) and an inert electrode (graphite), both immersed in a solution containing $\text{Na}_2\text{S}_2\text{O}_8(\text{aq})$ and $\text{Na}_2\text{SO}_4(\text{aq})$ and connected by a copper wire. According to Reaction 1, the presence of $\text{S}_2\text{O}_8^{2-}(\text{aq})$ initiates the oxidation of $\text{Fe}^{2+}(\text{aq})$ at the iron electrode to form $\text{Fe}^{3+}(\text{aq})$ and generate the electrons that flow through the copper wire to the graphite electrode, where $\text{Fe}^{3+}(\text{aq})$ is reduced back to $\text{Fe}^{2+}(\text{aq})$.

Therefore, not adding the $\text{Na}_2\text{S}_2\text{O}_8(\text{aq})$ would result in a voltmeter reading of 0 V because without $\text{S}_2\text{O}_8^{2-}(\text{aq})$, $\text{Fe}^{3+}(\text{aq})$ cannot form and its reduction at the cathode cannot occur, resulting in no current flowing through the cell and no cell potential.

(Choice B) $\text{FeCl}_3(\text{aq})$ is a source of $\text{Fe}^{3+}(\text{aq})$, which would be reduced at the cathode, resulting in current flowing through the cell with a nonzero cell potential.

(Choice C) Platinum is an electrically conductive metal and would still allow current to flow through the cell (ie, $E^\circ_{\text{cell}} \neq 0 \text{ V}$).

(Choice D) $\text{Na}_2\text{SO}_4(\text{aq})$ and $\text{MgSO}_4(\text{aq})$ are both strong **electrolytes** that would act as charge carriers to facilitate charge migration and enable the flow of electrons in the cell.

Educational objective:

The standard cell potential E°_{cell} is the voltage difference between the anode and cathode. If no current is flowing through the cell, E°_{cell} is 0 V.

Time spent: 0 sec
Subject:
General Chemistry

QID: 403859
System:
Solutions and Electrochemistry

Researchers investigated the role of an electrochemically enhanced sulfate radical-based system in the inactivation of antibiotic-resistant *Escherichia coli* bacteria. An electrochemical cell (Figure 1) was constructed using an iron electrode and a graphite electrode connected by a copper wire joined to a voltmeter. Both electrodes were placed in a beaker containing a 3 mM solution of $\text{Na}_2\text{SO}_4(\text{aq})$ and a 50 mM solution of a peroxydisulfate salt, $\text{Na}_2\text{S}_2\text{O}_8(\text{aq})$. During the operation of the cell, $\text{Fe}^{2+}(\text{aq})$ was formed at the iron electrode and subsequently oxidized to $\text{Fe}^{3+}(\text{aq})$ by a reaction with $\text{S}_2\text{O}_8^{2-}(\text{aq})$. At the graphite electrode, $\text{Fe}^{2+}(\text{aq})$ was regenerated.

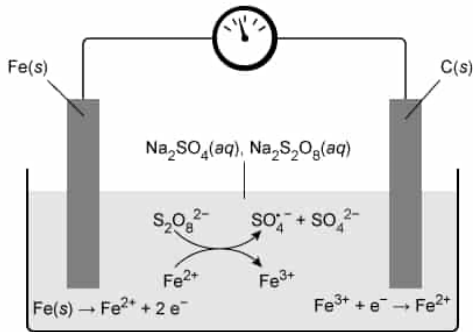
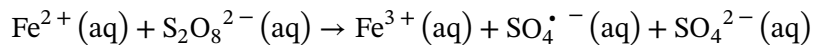


Figure 1 Electrochemical cell utilized in the inactivation of antibiotic-resistant *E. coli*

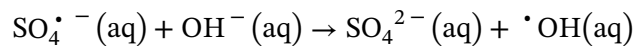
The reactions in the cell also produced both sulfate and hydroxyl radicals (ie, species with an unpaired electron). Sulfate radicals ($\text{SO}_4^{\bullet-}$) inactivate *E. coli* through membrane destruction and were produced in the electrochemical cell by Reaction 1.



Reaction 1

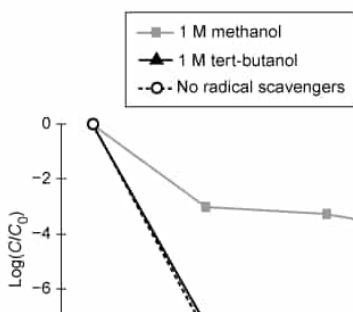
Operation of the cell generated enough energy to cause the electrolysis of water in which water was oxidized at one electrode to form $\text{H}^+(\text{aq})$ and reduced at the other electrode to form $\text{OH}^-(\text{aq})$.

Consequently, some of the available $\text{OH}^-(\text{aq})$ reacted with $\text{SO}_4^{\bullet-}(\text{aq})$ to form hydroxyl radicals ($\text{OH}^{\bullet}$), which also aided in the inactivation of *E. coli*.



Reaction 2

To determine which radical species predominately inactivated *E. coli* in the cell, two radical scavengers, methanol and *tert*-butanol, were used to evaluate the contributions of $\text{OH}^{\bullet}(\text{aq})$ and $\text{SO}_4^{\bullet-}(\text{aq})$. Methanol consumes both $\text{OH}^{\bullet}(\text{aq})$ and $\text{SO}_4^{\bullet-}(\text{aq})$ whereas *tert*-butanol primarily consumes $\text{OH}^{\bullet}(\text{aq})$. The researchers performed a series of experiments in which the concentration of *E. coli* was monitored in the active cell with and without the radical scavengers as a function of time. The results are shown in Figure 2.



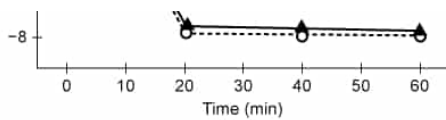
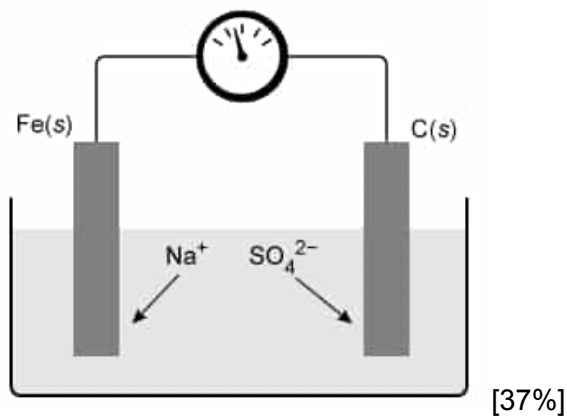


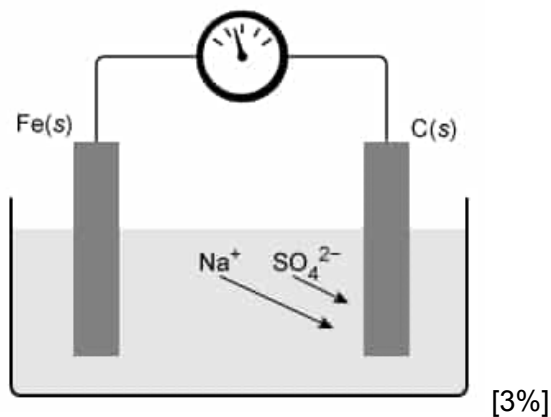
Figure 2 The effect of radical scavengers on the inactivation of antibiotic-resistant *E. coli* (Note: C_0 is the initial concentration of *E. coli*, and C is the concentration of *E. coli* at a specific time during the experiment.)

Which of the following diagrams illustrates the migration of the Na^+ and SO_4^{2-} ions in the electrochemical cell as it operates?

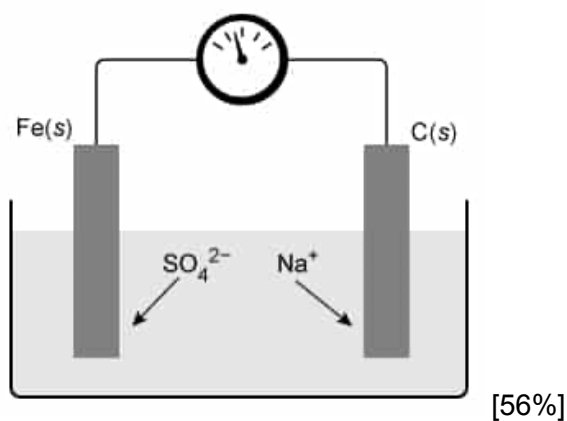
☐ A.



☐ B.

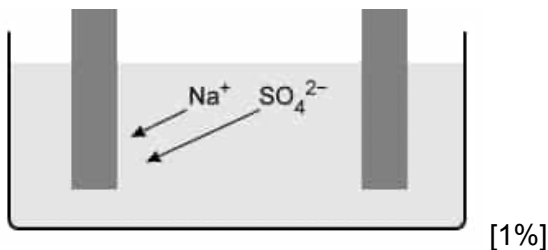


✓ ☒ C.



☐ D.

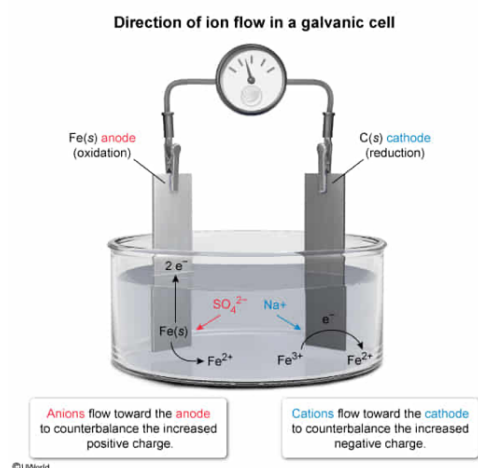




Correct

57% answered correctly

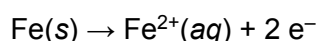
Explanation:



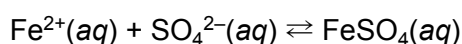
In a **galvanic cell**, an **oxidation-reduction** (redox) reaction occurs **spontaneously** (ie, an external power source is not needed to drive the reaction). This reaction produces electrons that travel from the **anode** (ie, the electrode where oxidation occurs) to the **cathode** (ie, the electrode where reduction occurs).

Between the oxidation and reduction components of an electrochemical cell, an **ion charge balance** is maintained by a **conductive electrolyte** that allows for the **migration** of ions within a single-cell solution or within a **salt bridge** joining two half-cell solutions.

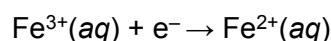
During operation of the electrochemical cell described in the passage (ie, a galvanic cell), Fe(s) atoms are oxidized to Fe²⁺(aq):



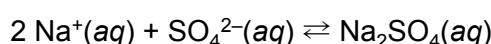
As more Fe²⁺(aq) is formed, the positive charges are counterbalanced by negatively charged SO₄²⁻ ions migrating within the cell solution toward the Fe(s) electrode:



At the same time, Fe³⁺(aq) (formed via Reaction 1) is reduced at the surface of the C(s) (graphite) electrode:



As more Fe³⁺(aq) is consumed, the increasing negative charges from the remaining SO₄²⁻(aq) are counterbalanced by positively charged Na⁺(aq) migrating within the cell solution toward the C(s) electrode:



Therefore, SO₄²⁻(aq) migrates toward the Fe(s) electrode and Na⁺(aq) migrates toward the C(s) electrode,

as depicted in the diagram presented in Choice C.

(Choices A, B, and D) Negatively charged ions (eg, SO_4^{2-}) migrate toward the Fe(s) anode (ie, where oxidation occurs and positive charges increase), and positively charged ions (eg, K^+) migrate toward the C(s) cathode (ie, where reduction occurs and negative charges increase).

Educational objective:

As a galvanic cell operates, ions from a conductive electrolyte within the cell solution (or within an adjoining salt bridge between half-cells) maintains a charge balance between the oxidation and reduction components of the cell. Anions migrate toward the anode, where oxidation occurs, and cations migrate toward the cathode, where reduction occurs.

Time spent:0 sec

Subject:
General Chemistry

QID:403860

System:
Solutions and Electrochemistry

Researchers investigated the role of an electrochemically enhanced sulfate radical-based system in the inactivation of antibiotic-resistant *Escherichia coli* bacteria. An electrochemical cell (Figure 1) was constructed using an iron electrode and a graphite electrode connected by a copper wire joined to a voltmeter. Both electrodes were placed in a beaker containing a 3 mM solution of $\text{Na}_2\text{SO}_4(\text{aq})$ and a 50 mM solution of a peroxydisulfate salt, $\text{Na}_2\text{S}_2\text{O}_8(\text{aq})$. During the operation of the cell, $\text{Fe}^{2+}(\text{aq})$ was formed at the iron electrode and subsequently oxidized to $\text{Fe}^{3+}(\text{aq})$ by a reaction with $\text{S}_2\text{O}_8^{2-}(\text{aq})$. At the graphite electrode, $\text{Fe}^{2+}(\text{aq})$ was regenerated.

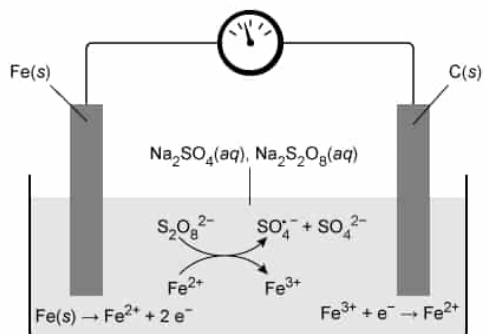
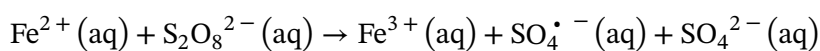


Figure 1 Electrochemical cell utilized in the inactivation of antibiotic-resistant *E. coli*

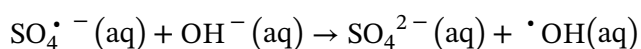
The reactions in the cell also produced both sulfate and hydroxyl radicals (ie, species with an unpaired electron). Sulfate radicals ($\text{SO}_4^{\bullet -}$) inactivate *E. coli* through membrane destruction and were produced in the electrochemical cell by Reaction 1.



Reaction 1

Operation of the cell generated enough energy to cause the electrolysis of water in which water was oxidized at one electrode to form $\text{H}^+(\text{aq})$ and reduced at the other electrode to form $\text{OH}^-(\text{aq})$.

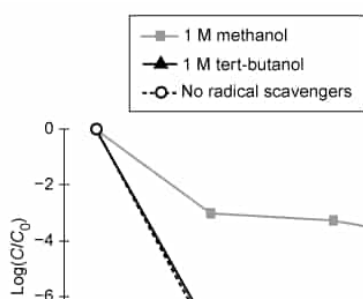
Consequently, some of the available $\text{OH}^-(\text{aq})$ reacted with $\text{SO}_4^{\bullet -}(\text{aq})$ to form hydroxyl radicals ($\bullet\text{OH}$), which also aided in the inactivation of *E. coli*.



Reaction 2

To determine which radical species predominately inactivated *E. coli* in the cell, two radical scavengers, methanol and *tert*-butanol, were used to evaluate the contributions of $\bullet\text{OH}(\text{aq})$ and $\text{SO}_4^{\bullet -}(\text{aq})$.

Methanol consumes both $\bullet\text{OH}(\text{aq})$ and $\text{SO}_4^{\bullet -}(\text{aq})$ whereas *tert*-butanol primarily consumes $\bullet\text{OH}(\text{aq})$. The researchers performed a series of experiments in which the concentration of *E. coli* was monitored in the active cell with and without the radical scavengers as a function of time. The results are shown in Figure 2.



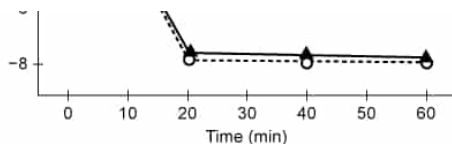


Figure 2 The effect of radical scavengers on the inactivation of antibiotic-resistant *E. coli* (Note: C_0 is the initial concentration of *E. coli*, and C is the concentration of *E. coli* at a specific time during the experiment.)

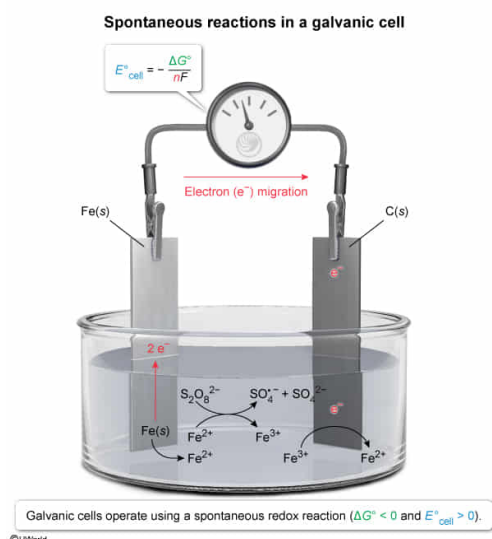
At 25 °C, the redox reaction occurring in the electrochemical cell has a:

- ✓ ☒ A. negative ΔG° and a positive E°_{cell} . [68%]
☐ B. negative ΔG° and a negative E°_{cell} . [6%]
☐ C. positive ΔG° and a negative E°_{cell} . [21%]
☐ D. positive ΔG° and a positive E°_{cell} . [2%]

Correct

69% answered correctly

Explanation:



In an electrochemical cell, an **oxidation-reduction** (redox) reaction converts chemical energy into

electrical energy (ie, a **galvanic cell**) or vice versa (ie, an **electrolytic cell**). The change in the **standard Gibbs free energy** ΔG° produced by these energy conversions represents the maximum amount of work that can be performed during the reaction. Consequently, ΔG° and the standard cell potential E°_{cell} (ie, the voltage difference between the anode and cathode) are related by:

$$\Delta G^\circ = -nFE^\circ_{\text{cell}}$$

or equivalently,

$$E^\circ_{\text{cell}} = -\frac{\Delta G^\circ}{nF}$$

where n is the number of electrons transferred per reaction and F is **Faraday's constant**.

In a galvanic cell, a **spontaneous** reaction occurs (ie, $\Delta G^\circ < 0$; $E^\circ_{\text{cell}} > 0$). In an electrolytic cell, an external power source is required to drive a **nonspontaneous** reaction (ie, $\Delta G^\circ > 0$; $E^\circ_{\text{cell}} < 0$).

The electrochemical cell described in the passage operates as a galvanic cell (ie, no external power source is used). Therefore, the redox reaction occurring in the given galvanic cell is spontaneous with a negative ΔG° and positive E°_{cell} .

(Choices B, C, and D) In a galvanic cell, a spontaneous redox reaction occurs with $\Delta G^\circ < 0$ and $E^\circ_{\text{cell}} > 0$.

Educational objective:

The standard Gibbs free energy ΔG° for a redox reaction is related to the number of electrons (n) transferred per reaction, the standard cell potential E°_{cell} , and Faraday's constant F , according to the equation $\Delta G^\circ = -nFE^\circ_{\text{cell}}$. For a spontaneous reaction, $\Delta G^\circ < 0$ and $E^\circ_{\text{cell}} > 0$; for a nonspontaneous reaction, $\Delta G^\circ > 0$ and $E^\circ_{\text{cell}} < 0$.

Time spent:0 sec

Subject:
General Chemistry

QID:403861

System:
Solutions and Electrochemistry

Researchers investigated the role of an electrochemically enhanced sulfate radical-based system in the inactivation of antibiotic-resistant *Escherichia coli* bacteria. An electrochemical cell (Figure 1) was constructed using an iron electrode and a graphite electrode connected by a copper wire joined to a voltmeter. Both electrodes were placed in a beaker containing a 3 mM solution of $\text{Na}_2\text{SO}_4(\text{aq})$ and a 50 mM solution of a peroxydisulfate salt, $\text{Na}_2\text{S}_2\text{O}_8(\text{aq})$. During the operation of the cell, $\text{Fe}^{2+}(\text{aq})$ was formed at the iron electrode and subsequently oxidized to $\text{Fe}^{3+}(\text{aq})$ by a reaction with $\text{S}_2\text{O}_8^{2-}(\text{aq})$. At the graphite electrode, $\text{Fe}^{2+}(\text{aq})$ was regenerated.

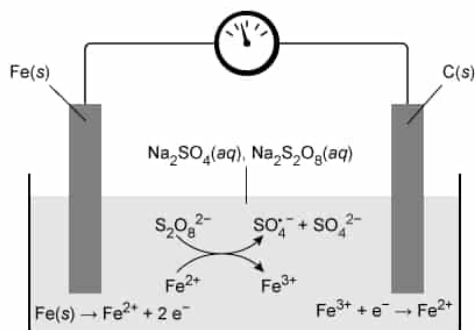
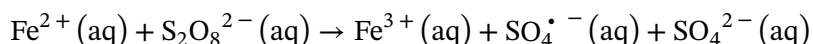


Figure 1 Electrochemical cell utilized in the inactivation of antibiotic-resistant *E. coli*

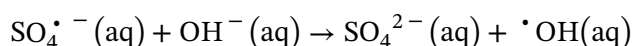
The reactions in the cell also produced both sulfate and hydroxyl radicals (ie, species with an unpaired electron). Sulfate radicals ($\text{SO}_4^{\bullet -}$) inactivate *E. coli* through membrane destruction and were produced in the electrochemical cell by Reaction 1.



Reaction 1

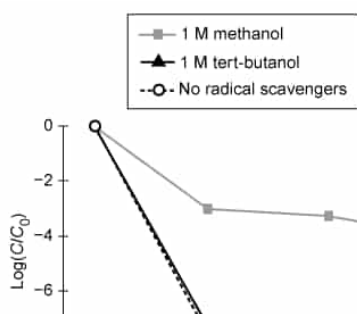
Operation of the cell generated enough energy to cause the electrolysis of water in which water was oxidized at one electrode to form $\text{H}^+(\text{aq})$ and reduced at the other electrode to form $\text{OH}^-(\text{aq})$.

Consequently, some of the available $\text{OH}^-(\text{aq})$ reacted with $\text{SO}_4^{\bullet -}(\text{aq})$ to form hydroxyl radicals ($\text{OH}^{\bullet}$), which also aided in the inactivation of *E. coli*.



Reaction 2

To determine which radical species predominately inactivated *E. coli* in the cell, two radical scavengers, methanol and *tert*-butanol, were used to evaluate the contributions of $\text{OH}^{\bullet}(\text{aq})$ and $\text{SO}_4^{\bullet -}(\text{aq})$. Methanol consumes both $\text{OH}^{\bullet}(\text{aq})$ and $\text{SO}_4^{\bullet -}(\text{aq})$ whereas *tert*-butanol primarily consumes $\text{OH}^{\bullet}(\text{aq})$. The researchers performed a series of experiments in which the concentration of *E. coli* was monitored in the active cell with and without the radical scavengers as a function of time. The results are shown in Figure 2.



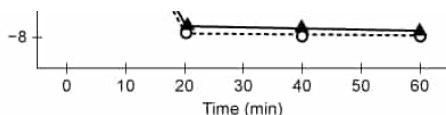


Figure 2 The effect of radical scavengers on the inactivation of antibiotic-resistant *E. coli* (Note: C_0 is the initial concentration of *E. coli*, and C is the concentration of *E. coli* at a specific time during the experiment.)

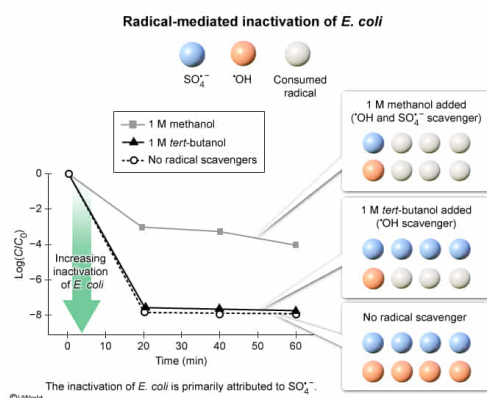
Based on Figure 2, what is the best conclusion regarding the role of the radical species involved in the inactivation of *E. coli*?

- ☐ A. Primarily $\cdot\text{OH}$ radicals contribute to the inactivation of *E. coli*. [27%]
- ✓ ☒ B. Primarily $\text{SO}_4^{\cdot-}$ radicals contribute to the inactivation of *E. coli*. [60%]
- ☐ C. Both $\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$ radicals contribute equally to the inactivation of *E. coli*. [9%]
- ☐ D. $\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$ radicals do not significantly contribute to the inactivation of *E. coli*. [2%]

Correct

60% answered correctly

Explanation:



According to the passage, antibiotic-resistant *E. coli* bacteria are inactivated by both $\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$ radicals. Because both radical species are formed in the electrochemical cell, the contribution of $\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$ to the inactivation of *E. coli* must be evaluated using two different radical scavengers: methanol (a scavenger of both $\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$) and *tert*-butanol (primarily an $\cdot\text{OH}$ scavenger).

In Figure 2, the logarithmic ratio of the *E. coli* concentration (C) at a specific time during the experiment and the initial *E. coli* concentration (C_0) is shown as a function of time. As the inactivation of *E. coli* increases, $\log(C/C_0)$ becomes increasingly negative (ie, $C < C_0$). When no radical scavengers are present in the cell, $\log(C/C_0) \approx -8$ (ie, inactivation is at a maximum).

When methanol is added, both $\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$ are consumed and $\log(C/C_0)$ becomes less negative (ie, $\log(C/C_0) \approx -4$) because fewer radicals are present in the solution (ie, less inactivated *E. coli*). However, when *tert*-butanol is added, $\cdot\text{OH}$ is consumed while the majority of $\text{SO}_4^{\cdot-}$ is left in the solution and $\log(C/C_0) \approx -8$ (ie, about the same as when no scavengers were present). Therefore, it can be concluded from Figure 2 that $\text{SO}_4^{\cdot-}$ must be the primary inactivator of *E. coli* in the electrochemical cell.

(Choices A and C) Because $\log(C/C_0) \approx -8$ when the cell is operating both with and without an $\cdot\text{OH}$ scavenger, primarily $\text{SO}_4^{\cdot-}$ must contribute to the inactivation of *E. coli*.

(Choice D) If both $\cdot\text{OH}$ and $\text{SO}_4^{\cdot-}$ did not contribute to the inactivation of *E. coli*, $\log(C/C_0)$ would be

equal to zero after 60 minutes, which is not the case.

Educational objective:

Experimental conclusions can be determined through graphical interpretation of data.

Time spent:0 sec

Subject:
General Chemistry

QID:403862

System:
Solutions and Electrochemistry